Leonardo Novicki Neto

Data Scientist | Machine Learning

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Data Scientist with an M.Sc. in Electrical Engineering (UFPR) and a B.Sc. in Physics, working on end-to-end Machine Learning — from identifying the business opportunity to deploying and maintaining models in production. Experience with regression, classification and time-series models applied to large volumes of sensor data, using Python, PyTorch and AWS. I enjoy turning complex operational problems into analytical solutions with measurable impact, working directly with business stakeholders.

TECHNICAL SKILLS

Languages	Python, SQL (MySQL, Amazon Athena), MATLAB
ML & Deep Learning	PyTorch, PyTorch Lightning, ConvLSTM, ConvGRU, CNN+Attention, LSTM, GRU, XGBoost, Random Forest, regression, time series, clustering, scikit-learn
Computer Vision	YOLO, YOLOP, OpenCV, semantic segmentation, genetic algorithms
Cloud & MLOps	AWS (Athena, S3, Glue, Lambda, SageMaker, ECS), Kedro, MLflow, Docker, REST APIs, awswrangler
Data & BI	Pandas, Polars, ETL, Power BI
Other	Git, ROS (Robot Operating System), Google Protocol Buffers, Agile (daily, sprint planning, sprint review)

PROFESSIONAL EXPERIENCE

Rumo Logística
Aug 2024 – Aug 2026

Data Scientist — Data Intelligence
Curitiba, Brazil

Joined through the Talento Inovação programme (IEL) for master's graduates and was hired directly by the company in April 2025.

- Raised precision from **55% to 95%** and recall from **39% to 60%** in predicting track closures caused by gauge widening, against a linear-regression baseline — increasing the accuracy of preventive maintenance and avoiding the high operational cost of track closures.
- Led the Data Science workstream of VIAbiliza, a project **identified by McKinsey as one of the highest-impact initiatives in the company**. I proposed, developed and implemented end-to-end the predictive models currently running in production.
- Identified the state of the art in the literature for predicting track-geometry degradation (LSTM with spatial convolution, applied to high-speed railways in Tokyo) and adapted the solution to a heavy-haul freight context, with custom preprocessing, training and evaluation criteria.
- Developed and trained deep learning models for spatio-temporal forecasting of track irregularities — ConvLSTM, ConvGRU and CNN+Attention architectures in PyTorch / PyTorch Lightning — with a custom exogenous-variable fusion mechanism (embeddings + concatenation along the channel axis).
- Solved the geographic misalignment between runs of the track geometry car — a well-known problem in the industry with no prior solution at the company — with an algorithm that aligns **over 99%** of the data, enabling all subsequent predictive modelling.
- Designed and implemented the data and ML pipeline in Kedro: from ETL on Amazon Athena to dataset generation on S3, training with experiment tracking in MLflow, and production inference — parameterised by target variable and able to compare multiple candidate models.
- Deployment and maintenance of models in production: batch inference through scheduled pipelines and a containerised service on AWS ECS consumed by other teams, integrated via REST APIs.
- Processed track-geometry data at scale — measurements every **25 cm across more than 1,500 km**, with runs every two weeks in both directions: cleaning, normalisation and construction of the spatio-temporal windows used as model input.
- First project: a sleeper (tie) maintenance prioritiser, built end-to-end in around 5 months — ETL, enrichment with operational data (gross tonnage and maintenance history), a predictive model of gauge degradation, and a Power BI dashboard used in strategic planning for the Rondonópolis–Santos corridor.
- Feature engineering from multiple sources — gross tonnage transported (MGT), infrastructure defects, peak and degradation indicators, vehicle-track interaction (VTI), rainfall, seasonality and sleeper type — transformed into tensors and fed to the models as exogenous variables.
- Helped build the ingestion services that began feeding AWS Athena: the track geometry car data existed only as .csv files, which made any large-scale analysis unfeasible.

- Ran the comparative evaluation between linear regression, tree-based models (Random Forest, XGBoost) and recurrent architectures.
- Feasibility analysis of a new onboard sensor on the track geometry car for rail-lift detection, producing the insights that guided the installation decision.
- AWS stack: ingestion and querying on Amazon Athena (awswrangler), processing with AWS Glue and Lambda, storage on S3, development and training on Amazon SageMaker and execution on ECS; handling of large data volumes with Pandas and Polars.
- Technical documentation of projects and models, working collaboratively with business (Track Reliability), technology and data teams in an agile routine of daily, sprint planning and sprint review.

CARISSMA Institute of Automated Driving

Sep 2022 – Mar 2023

Researcher — German government scholarship

Ingolstadt, Germany

- Development of the TWICE Dataset, a public dataset for autonomous vehicle research with camera, radar, LiDAR and GNSS/IMU sensors, under the supervision of Professors Werner Huber and Eduardo Parente Ribeiro. Published in a peer-reviewed international journal.
- Responsible for labelling the entire dataset, done semi-automatically by combining computer vision algorithms (YOLO) with a genetic algorithm.
- Analysis and validation using semantic segmentation (YOLOP) to quantify the simulation-to-reality gap between the real tests and their digital twins.
- Processing, transformation and organisation of the full volume of data collected by the institute's researchers, publishing it in an agnostic and accessible format and developing the tools to open, visualise and process the files. Python as the main tool, alongside MATLAB, ROS and Google Protocol Buffers.

Santa Lúcia Materiais de Construção

Jan 2016 – Sep 2022 · Nov 2023 – Aug 2024

IT Technician | Customer Service

Curitiba, Brazil

- Built an asset management system with a SQL database (MySQL) and a graphical interface in Python (Tkinter), with screens to populate the database and generate reports.
- Installation and maintenance of the computer network and electronic security systems, alongside customer service and administrative support.

EDUCATION

M.Sc. in Electrical Engineering

Mar 2022 – Nov 2023

Federal University of Paraná (UFPR), Brazil

Thesis: *TWICE Dataset: Digital Twin of Test Scenarios in a Controlled Environment*. Coursework: Applied Artificial Intelligence, Machine Learning, Computer Vision, Signals and Systems, Advanced Methods in Electronic Systems.

Visiting Master's Researcher — Automated Driving

Sep 2022 – Mar 2023

Technische Hochschule Ingolstadt (THI), Germany

German government scholarship to carry out the master's research project at CARISSMA.

B.Sc. in Physics

Aug 2016 – Aug 2021

Federal University of Paraná (UFPR), Brazil

Graduated with a mean grade of 82.17 (0–100 scale), never failing a course. Undergraduate research (CNPq scholarship) with Prof. Dr. Cyro Ketzer Saul on the development of a supercapacitor — data collection, analysis and processing.

PUBLICATIONS

- *TWICE Dataset: Digital Twin of Test Scenarios in a Controlled Environment* — International Journal of Vehicle Systems Modelling and Testing (2025). doi.org/10.1504/IJVSMT.2025.147353
- Pre-print (arXiv) — arxiv.org/abs/2310.03895
- Project page — twicedataset.github.io/site
- Full thesis — UFPR Digital Archive acervodigital.ufpr.br/handle/1884/87239

ADDITIONAL TRAINING

- Machine Learning Specialization — Stanford University / DeepLearning.AI (Feb – Mar 2024)

LANGUAGES

Portuguese	Native
English	Advanced — master's thesis written and presented in English; 6 months in Germany with daily communication in the language
Spanish	Intermediate